

# TrES-2b

R-1308

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-2\_130503 · created 2026-07-31T18:09:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456415.89546 +/- 0.00022 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.117 +/- 0.016                   |
| Transit depth (Rp/Rs)^2                           | 1.38 +/- 0.38 [%]                 |
| Orbital Inclination (inc)                         | 83.71 +/- 0.46                    |
| Airmass coefficient 1 (a1)                        | 0.9895 +/- 0.0062                 |
| Airmass coefficient 2 (a2)                        | 0.0103 +/- 0.0097                 |
| Scatter in the residuals of the lightcurve fit is | 0.63 %                            |
| Best Comparison Star                              | #3 - [65, 277]                    |
| Optimal Aperture                                  | 4.16                              |
| Optimal Annulus                                   | 9.44                              |
| Transit Duration (day)                            | 0.0699 +/- 0.0086                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.117 ± 0.016 vs 0.1254 (deviation 0.53σ) |
| Duration fit vs published | 0.0699 d vs 0.0746 d (deviation 0.54σ)    |
| Mid-time O–C              | -0.5 min                                  |
| Depth SNR                 | 7.3                                       |
| Residual scatter          | 0.63 %                                    |

## Light curve

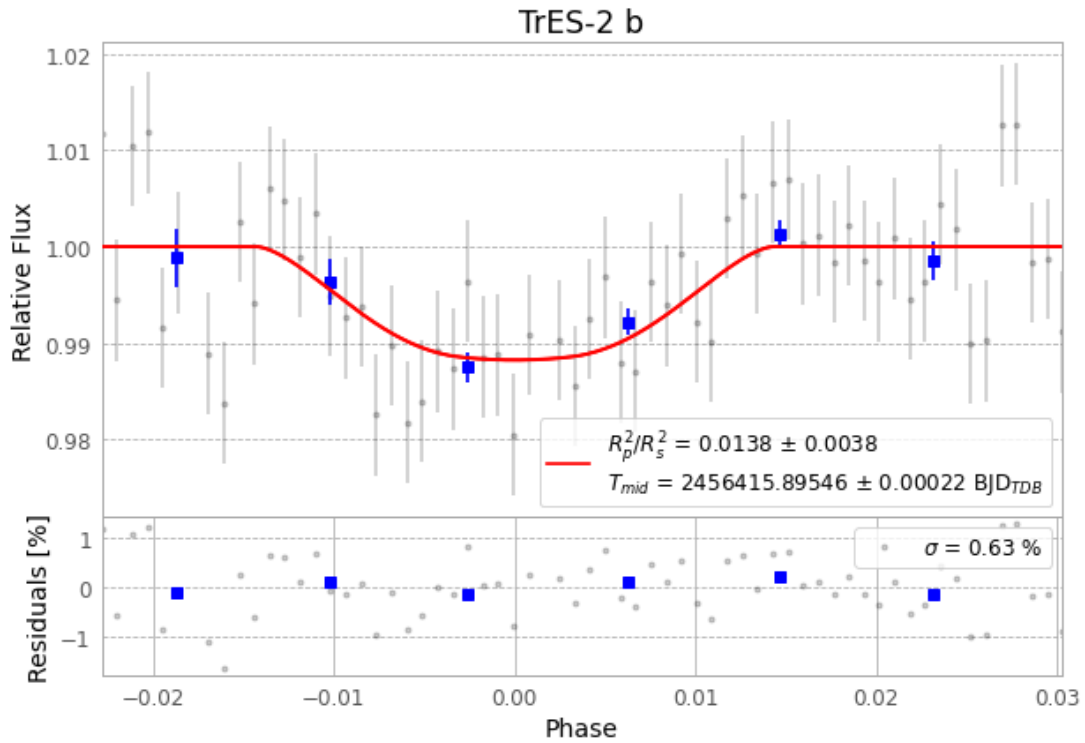


Figure 1 — FinalLightCurve\_TrES-2 b\_2013-05-03.png (SHA-256 b5c9d120de5f4dd1...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 234], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 b6d98d87175ec53c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

63 FITS frames (41 MB), TRES-2130503080612.FITS → TRES-2130503111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130503.log (156090dedaed...), FinalLightCurve\_TrES-2 b\_2013-05-03.png (b5c9d120de5f...), FinalLightCurve\_TrES-2 b\_2013-05-03.csv (f7bc9314748f...)

## Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 18:09Z.